

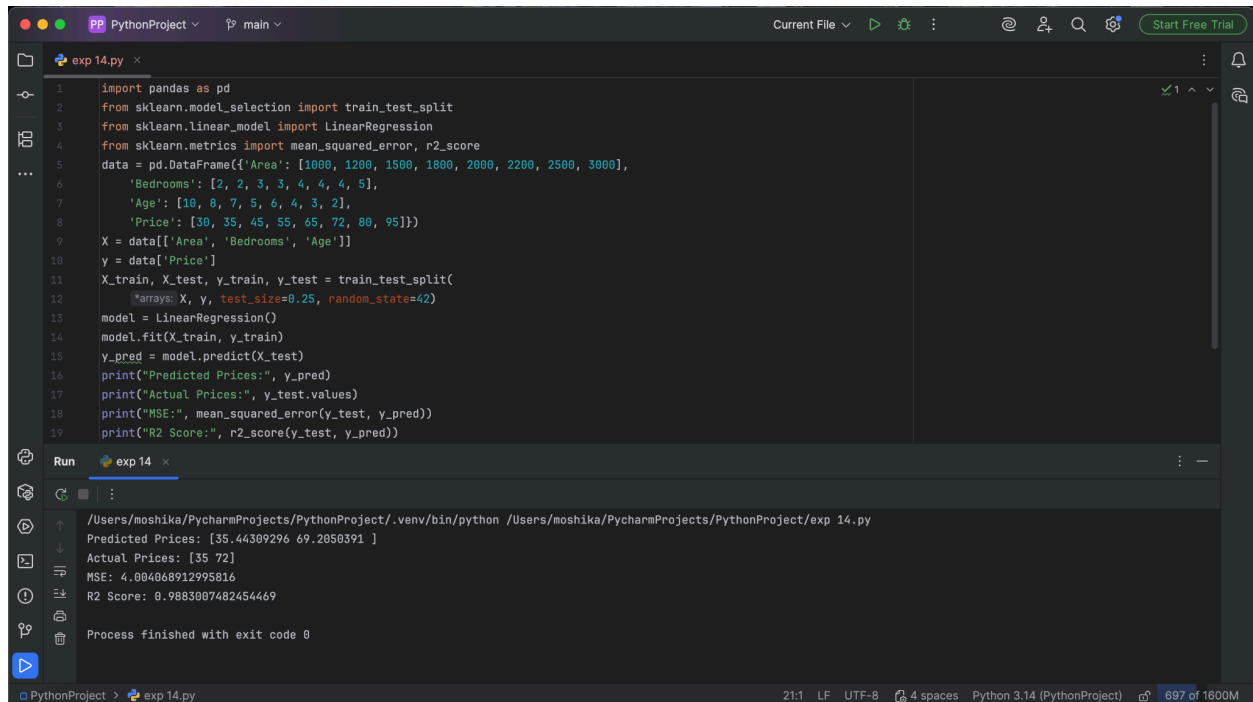
EXPERIMENT - 14

Aim: To predict house prices using an appropriate machine learning regression algorithm.

Procedure:

1. Load the house price dataset.
2. Preprocess the dataset.
3. Select relevant features such as area, bedrooms, and location.
4. Separate input features and target price.
5. Split the data into training and testing sets.
6. Train a regression model.
7. Predict house prices for test data.
8. Evaluate the model performance.
9. Display the predicted house prices.

Output:



The screenshot displays a Python IDE with a file named `exp 14.py`. The code imports `pandas`, `sklearn.model_selection` (for `train_test_split`), `sklearn.linear_model` (for `LinearRegression`), and `sklearn.metrics` (for `mean_squared_error` and `r2_score`). It creates a `DataFrame` with features: `Area`, `Bedrooms`, `Age`, and `Price`. The data is split into training and testing sets using `train_test_split` with `test_size=0.25` and `random_state=42`. A `LinearRegression` model is trained on the training data and used to predict prices for the test data. The output shows the predicted prices, actual prices, MSE, and R2 Score.

```
1 import pandas as pd
2 from sklearn.model_selection import train_test_split
3 from sklearn.linear_model import LinearRegression
4 from sklearn.metrics import mean_squared_error, r2_score
5 data = pd.DataFrame({'Area': [1800, 1200, 1500, 1800, 2000, 2200, 2500, 3000],
6                       'Bedrooms': [2, 2, 3, 3, 4, 4, 4, 5],
7                       'Age': [10, 8, 7, 5, 6, 4, 3, 2],
8                       'Price': [30, 35, 45, 55, 65, 72, 80, 95]})
9 X = data[['Area', 'Bedrooms', 'Age']]
10 y = data['Price']
11 X_train, X_test, y_train, y_test = train_test_split(
12     X, y, test_size=0.25, random_state=42)
13 model = LinearRegression()
14 model.fit(X_train, y_train)
15 y_pred = model.predict(X_test)
16 print("Predicted Prices:", y_pred)
17 print("Actual Prices:", y_test.values)
18 print("MSE:", mean_squared_error(y_test, y_pred))
19 print("R2 Score:", r2_score(y_test, y_pred))
```

Run exp 14

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 14.py
Predicted Prices: [35.44389296 69.2858391 ]
Actual Prices: [35 72]
MSE: 4.084868912995816
R2 Score: 0.9883887482454469
Process finished with exit code 0
```

Result: The machine learning model successfully predicted house prices based on the given features.